Inoetia Sasa

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Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY
FORMATIVE ASSESSMENT

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About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment Sasa

Learner: Moshibudi Inocetia	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Moshibudi Inocetia Sasa

Learner's signature: _____ Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Moshibudi Innocentia Sasa	Signature
Facilitator Name:	Signature:

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction b) Describe with the aid of diagrams, the configuration of motor connections c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	5 5 10

(a) 1) Induction Motor: Standard industrial motor where the rotor runs slightly slower than the stator field.

* Stator: Laminated iron core with windings that create a rotating magnetic field.

* Rotor: either a squirrel cage or wound rotor

2) ~~Synchor~~ Synchronous motor: operates at the exact speed of the magnetic field.

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a rotating magnetic field (RMF).
* This rotating field cuts across the rotor conductor and induces a current in the rotor (electromagnetic induction)
* The induction induced current creates it's own magnetic field in the rotor.

b) 1) Insulation resistance test
* Done using megger, measures resistance between windings and earth, high resistance = good insulation.

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2). Continuity test
* Checks if the windings are complete (not broken), done using a multimeter, ensures current can flow through all coils.

c) 1) Overload relay
* protects against excess current over a long time.

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2) Circuit breaker
* Automatically disconnects power during short circuits and overcurrent

2) Fuses
* Melt when current is too high, provides a quick protection against short circuits.

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4) Earth Leakage protection
* Detects current leaking to ground through damaged insulation and trips the circuit to prevent electric shock.

5) under-voltage relay → prevents the motor from drawing high currents and overheating.

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LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) List the different types of direct current motors and describe their construction	2
	b) Describe the applications of different types of direct current motors	2
	c) Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits.	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits.	2
	e) Calculate direct current motor properties	2

2) Types of DC motors

1) series motor

2) Shunt motor

3) Compound motor

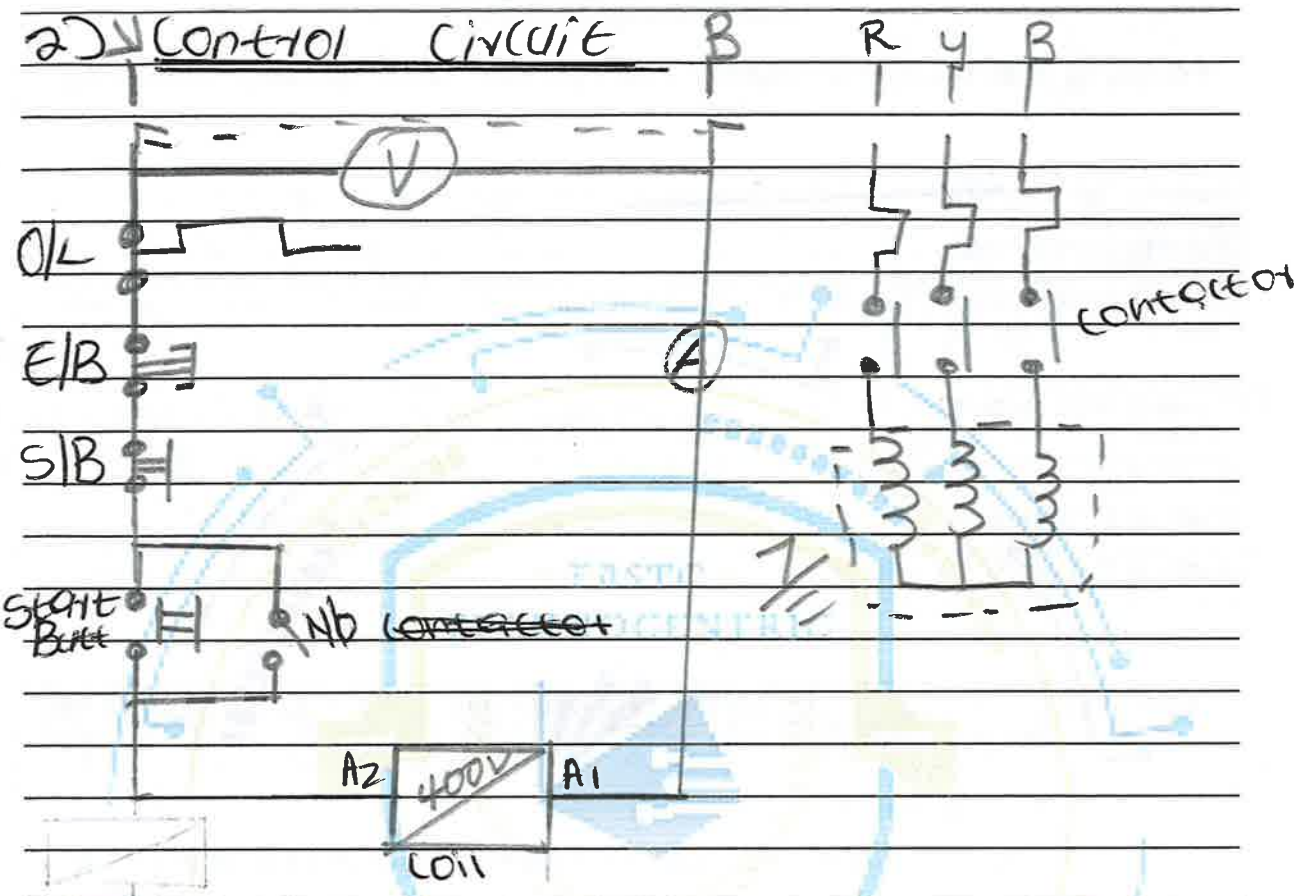
Construction of all DC motors

1) Stator produces magnetic field

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2) Control Circuit



d) D Time delay

* Allows temporary high current
* short delay: quick trips
* Long delay: avoids unnecessary tripping

2) Current rating

* Set according to motor
* too low -> Nuisance tripping
* too high -> motor may overheat and get damaged.

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cuts the stationery conductors (stator)
* This induces an alternating voltage (AC)
* Output changes direction continuously
-> Ac supply
> Generator (DC generator)
Principle:
* Also based on electromagnetic induction
Explanation:
* Armature rotates in a magnetic field
* induced voltage is converted to DC using a commutator
* Output is indirectional (DC)
b) > Alternators (AC)
* power stations -> generates electricity for the grid
* Vehicle alternators -> charge batteries
* Industrial plants -> continuous power supply
> Generator (DC)
* Batter charging systems
* Electroplating
* Welding machines
c) 1) Overcurrent protection
* protects against excess current
* uses circuit breakers or relays
2) Overload protection
* prevents overheating due to long-term high current
* uses thermal overload relays
3) Earth Fault protection
* Compares incoming and outgoing current
* Trips when there is a difference

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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0301 Construction of alternators and generators			✓
2	KT0302 Principle of operation of alternators and generators			✓
3	KT0303 Configuration of alternator and generator connections			✓
4	KT0304 Types and application of alternators and generators			✓
5	KT0305 Testing principles of alternators and generators			✓
6	KT0306 Protection of alternators and generators			✓
7	KT0307 Calculate properties of alternators and generators			✓

Learner Signature **Date**

Facilitators' Signature **Date**

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DECLARATION BY LEARNER			
I, Moshibudi Innocentia Sasa declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
SasaMP			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

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Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

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5. Complete learner toolkit

6. Complete assessor toolkit